MICROECONOMICS

MICRO-21

Time: 6 minutes

Outcome: Complete short-run cost schedules using total and average cost identities

Difficulty: ●●○ Intermediate

Cost Components & Cost Schedules

THE CORE IDEA

In the short run, total fixed cost (TFC) does not change with output, while total variable cost (TVC) rises or falls as production changes. Total cost combines both components: TC = TFC + TVC. For positive output Q, divide each total by Q: AFC = TFC / Q, AVC = TVC / Q, and ATC = TC / Q = AFC + AVC. At zero output, TVC is normally zero, TC equals TFC, and average costs are undefined because division by zero is impossible. On a total-cost graph, the vertical distance between TC and TVC equals TFC at every quantity.

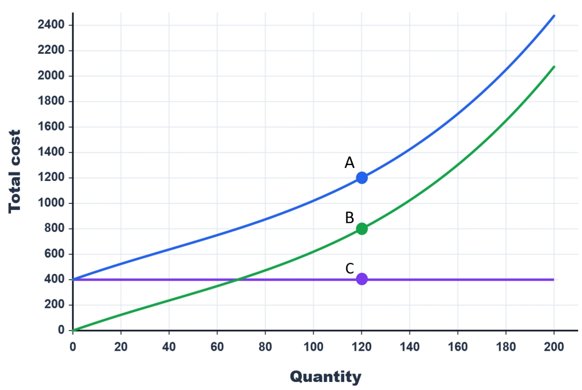
HOW TO RECOGNIZE IT

- A cost-schedule row is missing TFC, TVC, or TC.
- A fixed payment remains even when current output is zero.
- A cost changes when labor, materials, or production changes.
- The vertical gap between TC and TVC is compared across quantities.
- Total costs must be converted to per-unit AFC, AVC, or ATC.

WATCH OUT

Do not subtract or multiply fixed and variable cost when finding total cost: add them. Also distinguish total fixed cost from average fixed cost. TFC stays constant as output changes, but AFC = TFC / Q falls as the same fixed amount is spread across more units. Average costs are not defined at Q = 0.

WORKED EXAMPLE: BUILDING A COST SCHEDULE



At Q = 120, the graph shows total cost of $1,200 and total variable cost of $800. The $400 vertical gap between the two curves is total fixed cost because TC = TFC + TVC. Dividing each cost by 120 gives AFC = $3.33, AVC = $6.67, and ATC = $10.00. As a check, AFC + AVC = ATC.

CHECK YOURSELF

At Q = 10, a schedule shows TFC = $90 and TC = $250. What are TVC, AFC, AVC, and ATC, and how can you verify that AFC + AVC = ATC?

READY? Return to the game and master this concept.